

Dataset guide for CLIP quantization on multimodal entity linking

Three of the four requested datasets — WikiDiverse, MS COCO, and WNED-WIKI — are well-documented and publicly available; "WiMBo" does not appear to exist as a published benchmark.

WikiDiverse and MS COCO provide multimodal image-text data ideal for CLIP-based experiments, while WNED-WIKI is a text-only entity linking evaluation set. All three confirmed datasets fit comfortably on a single 4090 GPU setup, with COCO being the largest at ~19 GB of images. Below is everything needed to download, load, and preprocess each dataset.

WikiDiverse: multimodal entity linking from Wikinews

Paper: Wang et al., "WikiDiverse: A Multimodal Entity Linking Dataset with Diversified Contextual Topics and Entity Types," ACL 2022 (Long Papers), pp. 4785–4797. ([aclanthology](#)) ([ACL Anthology](#)) Available at [arXiv:2204.06347](#) and [ACL Anthology](#).

Official repo: github.com/wangxw5/wikiDiverse ([GitHub](#)) — contains evaluation scripts and Google Drive download links for all data files. ([Papers with Code](#))

Download links (Google Drive):

- Annotated data (train/val/test JSON)
- Data with 10 retrieved candidates per mention
- Entity descriptions ([entity2desc](#))
- Prior probability data ([P\(e|m\)](#))
- **Wikinews images** (pre-downloaded and cleaned, zipped)
- Entity-to-image-URL mappings for Wikipedia entities

HuggingFace Hub: Not available. No [datasets.load_dataset\(...\)](#) option exists — data must be downloaded manually from Google Drive.

Data format and structure: WikiDiverse uses JSON (Python lists of lists). At the **passage level**, each instance contains: [\[sentence_text, image_url, topic_category, \[\[mention_text, entity_type, start_char, end_char, wikipedia_url\], ...\]\]](#). ([github](#)) At the **mention level** (candidate data), each instance has 11 fields including left/right context tokens, ground-truth entity URL, candidate entity list, and topic label. ([GitHub](#)) Entity types include Person, Organization, Location, Event, and Other. ([OpenReview](#)) Images are stored as downloaded files, with paths reconstructable via MD5 hashing of the original Wikimedia filename using a provided Python snippet.

Scale: ~8K image-caption pairs and ([GitHub](#)) ~16.3K entity mentions ([arXiv](#)) across 10 topical domains (sports, politics, entertainment, etc.). The split is 6,312 / 755 / 757 passages for train/val/test, yielding 13,205 / 1,552 / 1,570 mention-level instances. ([GitHub](#)) The knowledge base covers ~16M Wikipedia entities.

([ACL Anthology](#))

Preprocessing: The repo provides [eval_cands_recall.py](#) for candidate recall evaluation and a Python snippet for image-path reconstruction using [hashlib.md5](#). Community implementations like [GEMEL](#) [\(GitHub\)](#) and [UniMEL](#) [\(GitHub\)](#) include preprocessed WikiDiverse data and loading code.

License: Dataset is **CC BY-SA 4.0**; [\(GitHub\)](#) code is Apache-2.0. [\(github\)](#)

MS COCO: the standard retrieval benchmark for CLIP evaluation

Papers: Lin et al., "Microsoft COCO: Common Objects in Context," [\(TensorFlow\)](#) ECCV 2014 ([arXiv:1405.0312](#)). For retrieval evaluation, the standard split comes from Karpathy & Li, "Deep Visual-Semantic Alignments for Generating Image Descriptions," CVPR 2015. [\(GitHub\)](#)

Official download ([cocodataset.org](#)):

Resource	Size	URL
2014 Train images (83K)	13 GB	http://images.cocodataset.org/zips/train2014.zip
2014 Val images (41K)	6 GB	http://images.cocodataset.org/zips/val2014.zip
2017 Train images (118K)	18 GB	http://images.cocodataset.org/zips/train2017.zip
2017 Val images (5K)	1 GB	http://images.cocodataset.org/zips/val2017.zip
2014 Annotations	241 MB	http://images.cocodataset.org/annotations/annotations_trainval2014.zip
2017 Annotations	241 MB	http://images.cocodataset.org/annotations/annotations_trainval2017.zip

An important detail: **2014 and 2017 contain the same images**, just reshuffled into different train/val splits.

[\(TensorFlow\)](#) For CLIP retrieval evaluation, you typically need the 2014 images plus the Karpathy split file.

Karpathy split file ([dataset_coco.json](#)): Available from a [GitHub mirror](#) [\(GitHub\)](#) or [Kaggle](#). This split re-partitions COCO 2014 into **113,287 train / 5,000 val / 5,000 test / 30,504 restval** images. Each image has 5 captions, giving **25,010 captions** in the 5K test set [\(GitHub\)](#) used for standard I2T/T2I recall reporting.

HuggingFace Hub options:

```
python
```

```
# Option 1: COCO with Karpathy splits (recommended for retrieval)
```

```
from datasets import load_dataset
```

```
ds = load_dataset("HuggingFaceM4/COCO") # includes Karpathy splits
```

```
# Option 2: Karpathy-specific
```

```
ds = load_dataset("yerevann/coco-karpathy") # 123K rows, parquet format
```

```
# Option 3: Full COCO by year and task
```

```
ds = load_dataset("shunk031/MSCOCO", year=2017, coco_task="captions")
```

Annotation format: Caption JSONs follow this structure: `{"images": [{"id": int, "file_name": str, "height": int, "width": int, ...}], "annotations": [{"id": int, "image_id": int, "caption": str}]}`. [Immersive Limit](#) Each image has **5 captions** [Sjtu](#) from different annotators. [Immersive Limit](#) Image filenames follow `COCO_train2014_000000XXXXXX.jpg` (2014) or `000000XXXXXX.jpg` (2017).

Scale: Over **330K total images** [V7 Labs](#) with [Sjtu](#) **~616K captions**. For CLIP retrieval benchmarking, only the 5K Karpathy test set matters — CLIP ViT-B/32 zero-shot baselines on this set are $R@1=58.4$ / $R@5=81.5$ for I2T, and $R@1=37.8$ / $R@5=62.4$ for T2I. [GitHub](#)

CLIP preprocessing pipelines:

```
python
```

```
# OpenAI CLIP
```

```
import clip
```

```
model, preprocess = clip.load("ViT-B/32", device="cuda")
```

```
# HuggingFace Transformers
```

```
from transformers import CLIPModel, CLIPProcessor
```

```
model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32")
```

```
processor = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")
```

```
# LAION CLIP Benchmark (standardized eval)
```

```
# pip install clip-benchmark
```

```
# clip_benchmark eval --dataset=mscoco_captions --task=zeroshot_retrieval \
```

```
# --model=ViT-B-32-quickgelu --pretrained=laion400m_e32 --batch_size=64
```

License: Annotations are **CC-BY 4.0**. [Hugging Face](#) [GitHub](#) Images are sourced from Flickr under mixed Creative Commons licenses [GitHub](#) — per-image license info is embedded in the annotation JSONs.

WiMBo does not appear to be a published dataset

Extensive searching across Google Scholar, arXiv, Semantic Scholar, Papers with Code, HuggingFace, and GitHub **returned no results for a dataset called "WiMBo"** in the context of multimodal entity linking. The only "WiMBo" found is an unrelated digital art history project (Watermark Imaging Box). [IEEE Xplore](#) [Semantic Scholar](#) The name may be a misspelling, an internal/unreleased dataset, or a confusion with one of these established MEL benchmarks:

- **WikiMEL / Wikidata-MEL** (18,880 samples) — encyclopedia-based MEL [GitHub](#) from Zhou et al., CCKS 2021; available at github.com/seukgcode/MELBench [GitHub](#)
- **Weibo-MEL** (25,602 samples) — social media MEL from Chinese Weibo, [GitHub](#) same repo [GitHub](#)
- **Richpedia-MEL** (17,806 samples) — multimodal knowledge graph MEL, [GitHub](#) same repo [GitHub](#)
- **M3EL** (1,100 movies, 181K mentions, 45K images) — movie-based MEL from Gan et al. [Jingrug](#)

If "WiMBo" refers to a Wikipedia-based Multimodal Benchmark, WikiDiverse or WikiMEL are the closest matches. If you can share the paper or context where you encountered the name, that would help track down the actual dataset.

WNED-WIKI: a text-only entity disambiguation benchmark

Paper: Guo & Barbosa, "Robust Named Entity Disambiguation with Random Walks," [arXiv](#) [University of Alberta](#) *Semantic Web* 9(4):459–479, 2018 (extended version of their CIKM 2014 paper). DOI: [10.3233/SW-170273](https://doi.org/10.3233/SW-170273).

Official sources:

- **GitHub (system code):** github.com/U-Alberta/wned
- **Data download:** [Borealis Dataverse](#) (University of Alberta), DOI: [10.7939/DVN/10968](https://doi.org/10.7939/DVN/10968)
- **KILT distribution:** Also available through [Facebook's KILT benchmark](#) as pre-formatted JSONL files

HuggingFace Hub:

```
python

from datasets import load_dataset
ds = load_dataset("facebook/kilt_tasks", "wned")
# Returns 6,770 total rows: validation (3,396) + test (3,376)
```

WNED-WIKI is definitively text-only — no images, no multimodal data. The task is linking textual mention spans in Wikipedia-sourced documents to their correct Wikipedia entities.

Data format: In the original distribution, documents are plain text files accompanied by XML annotation files mapping mention spans to gold Wikipedia entities. [GitHub](#) In the **KILT JSONL format**, each line is a JSON object with fields: `id`, `input` (text containing the mention), `output` (with `answer` entity name and `provenance` including `wikipedia_id`, `title`, character offsets), and `meta` (with `left_context`, `mention`, `right_context`).

Scale: ~310 documents with ~6,374 entity mentions. [GitHub](#) The KILT-formatted files are approximately **12.9 MB** (dev) and **13.3 MB** (test). [GitHub](#) There is **no training split** — WNED-WIKI is designed as an out-of-domain evaluation set, typically used alongside models trained on AIDA-CoNLL. [arXiv](#)

Related datasets in the same family: WNED-CWEB (~11K mentions from ClueWeb), MSNBC (~656 mentions), AQUAINT (~50 docs), ACE2004 (~256 mentions), and AIDA-CoNLL (~35K mentions). Guo & Barbosa cleaned and standardized several of these for consistent evaluation.

License: KILT distribution is **MIT licensed**. [GitHub](#) Original Borealis data is openly accessible. Underlying text derives from Wikipedia (CC BY-SA).

Practical setup for CLIP quantization on a 4090

For your CLIP quantization experiments targeting multimodal entity linking, here is the recommended workflow on a 24 GB VRAM 4090:

Storage requirements: WikiDiverse images are modest (a few GB at most). COCO 2014 train+val requires ~19 GB of images. WNED-WIKI is negligible at ~26 MB. Total disk footprint should stay under **25–30 GB**.

Recommended loading approach for each dataset:

1. **WikiDiverse** — Download manually from Google Drive links in the GitHub repo. Use the MD5-based image path mapping. No HuggingFace loader exists, so write a custom `torch.utils.data.Dataset` class that parses the JSON and loads images via PIL.
2. **MS COCO (retrieval)** — Use `load_dataset("HuggingFaceM4/COCO")` for the Karpathy split, or download images from cocodataset.org and the Karpathy JSON from the GitHub mirror. For CLIP evaluation, the LAION `clip-benchmark` package provides a turnkey retrieval eval pipeline. [PyPI](#)
3. **WNED-WIKI** — Use `load_dataset("facebook/kilt_tasks", "wned")` for immediate access. Since it is text-only, it's relevant for the text-encoder side of CLIP quantization — measuring how well quantized text embeddings preserve entity disambiguation quality.

Quantization-specific notes: A ViT-B/32 CLIP model uses ~0.6 GB in FP32 and ~0.15 GB in INT8, leaving ample VRAM on a 4090 for large batch inference. For quantization, consider using `torch.quantization`, GPTQ (via `auto-gptq`), or the HuggingFace `optimum` library with ONNX Runtime. Batch sizes of 256–512 for COCO retrieval eval should be feasible in quantized mode.